

LUXEMBOURG AP, Luxembourg

WMO: 065900

Lat: 49.627N

Lon: 6.212E

Elev: 376

StdP: 96.89

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.7	(c) -5.9	(d) -12.7	(e) 1.3	(f) -4.6	(g) -10.0	(h) 1.7	(i) -3.6	(j) 12.3	(k) 7.9	(l) 10.9	(m) 7.3	(n) 3.0	(o) 50	(p) 0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.2	(c) 29.8	(d) 19.0	(e) 27.8	(f) 18.2	(g) 26.0	(h) 17.5	(i) 20.0	(j) 27.4	(k) 19.2	(l) 25.9	(m) 18.4	(n) 24.4	(o) 3.1	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	13.3	22.2	16.8	12.5	21.3	16.0	11.9	20.9	58.8	27.6	56.0	26.2	53.3	24.5	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.4	(b) 8.2	(c) 7.2	(e) -10.5	(f) 33.1	(g) 2.7	(h) 2.1	(i) -12.5	(j) 34.7	(k) -14.0	(l) 35.9	(m) -15.5	(n) 37.1	(o) -17.5	(p) 38.6		
(a) 9.4	(b) 8.2	(c) 7.2	(e) -11.1	(f) 21.6	(g) 2.8	(h) 0.9	(i) -13.1	(j) 22.3	(k) -14.7	(l) 22.8	(m) -16.3	(n) 23.3	(o) -18.3	(p) 24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	1.3	2.4	5.9	9.7	13.6	16.9	18.9	18.4	14.6	10.3	5.4	2.3
	DBStd	7.26	4.03	3.93	3.84	4.14	3.85	3.78	3.73	3.61	3.16	3.51	3.53	3.99
	HDD10.0	1106	269	212	136	56	10	1	0	0	2	39	142	240
	HDD18.3	3183	527	445	386	261	154	73	39	43	121	248	388	498
	CDD10.0	1112	0	0	8	46	121	206	277	261	138	49	4	1
	CDD18.3	147	0	0	0	1	7	28	57	46	7	0	0	0
	CDH23.3	1126	0	0	0	6	48	217	454	357	43	1	0	0
	CDH26.7	307	0	0	0	0	6	52	131	115	4	0	0	0
Wind	WSAvg	3.3	3.7	3.7	3.8	3.4	3.3	3.1	3.1	2.8	3.0	3.1	3.1	3.7
Precipitation	PrecAvg	834	72	59	62	54	75	74	72	74	67	74	78	85
	PrecMax	1200	198	155	138	176	190	191	197	201	146	203	155	232
	PrecMin	530	6	6	7	5	7	5	6	14	14	2	10	4
	PrecStd	162	40	37	33	35	40	42	39	41	35	45	32	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	12.9	18.6	24.3	27.4	31.0	32.9	33.7	27.0	21.5	15.0	11.6
		MCWB	9.8	8.1	10.4	14.2	18.1	20.0	19.8	20.0	17.7	16.0	12.8	10.2
	2%	DB	9.8	10.5	15.9	21.3	24.4	28.0	29.8	29.6	24.4	18.8	12.9	10.3
		MCWB	8.7	7.4	9.4	12.7	15.8	18.7	18.9	18.9	16.9	14.6	11.1	9.2
	5%	DB	8.2	8.9	13.5	18.8	22.5	25.8	27.8	27.1	22.0	16.8	11.5	9.1
		MCWB	7.1	7.1	8.6	11.7	14.9	17.7	18.2	18.0	16.0	13.8	10.2	8.1
	10%	DB	6.8	7.8	11.4	16.5	20.3	23.7	25.9	24.8	19.9	15.2	10.2	7.8
		MCWB	5.8	6.4	8.0	10.6	13.9	16.7	17.6	17.3	15.0	13.0	9.1	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	9.5	11.7	15.1	19.0	21.2	21.0	21.0	18.9	16.6	13.0	10.7
		MCDB	10.9	11.1	15.9	22.1	25.3	28.8	29.2	28.8	24.4	19.9	14.2	11.2
	2%	WB	8.7	8.2	10.4	13.5	17.3	19.6	20.0	19.7	17.7	15.4	11.5	9.3
		MCDB	9.7	9.6	14.0	19.8	22.6	25.9	27.3	27.3	22.8	18.0	12.5	10.1
	5%	WB	7.2	7.3	9.5	12.2	15.7	18.5	19.1	18.9	16.6	14.3	10.3	8.2
		MCDB	8.0	8.7	12.4	17.4	20.7	24.3	26.3	25.3	20.9	16.4	11.4	9.0
	10%	WB	5.9	6.3	8.4	11.1	14.5	17.4	18.2	18.0	15.6	13.2	9.2	7.1
		MCDB	6.7	7.7	11.0	15.6	19.2	22.5	24.5	23.5	19.2	14.9	10.2	7.8
Mean Daily Temperature Range	5% DB	MDBR	4.1	5.3	7.1	8.4	8.7	9.1	9.2	8.9	7.9	6.3	4.4	4.0
		MCDBR	5.2	6.9	10.2	11.2	11.3	11.7	11.9	12.1	10.8	8.3	5.5	4.6
		MCWBR	4.7	4.9	5.6	5.4	5.4	5.4	4.6	4.8	5.0	4.9	4.2	4.1
	5% WB	MCDBR	5.1	5.9	8.2	10.1	10.3	10.6	10.8	10.3	9.3	7.3	5.2	4.5
		MCWBR	4.7	4.6	5.2	5.2	5.3	5.5	4.6	4.8	4.9	4.6	4.3	4.2
Clear-Sky Solar Irradiance	taub	0.301	0.327	0.355	0.385	0.395	0.397	0.393	0.390	0.363	0.347	0.321	0.300	
	taud	2.447	2.407	2.340	2.280	2.281	2.299	2.300	2.339	2.395	2.437	2.452	2.438	
	Ebn at Noon	755	817	850	859	864	862	859	845	832	780	719	700	
	Edh at Noon	62	81	103	121	126	125	123	113	97	78	62	55	
All-Sky Solar Radiation	RadAvg	0.81	1.53	2.73	4.26	5.03	5.60	5.32	4.55	3.43	1.97	0.91	0.63	
	RadStd	0.12	0.28	0.41	0.54	0.53	0.50	0.55	0.48	0.40	0.18	0.10	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.48	N/A	+0.45	+0.49	N/A	N/A	+64	N/A
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page